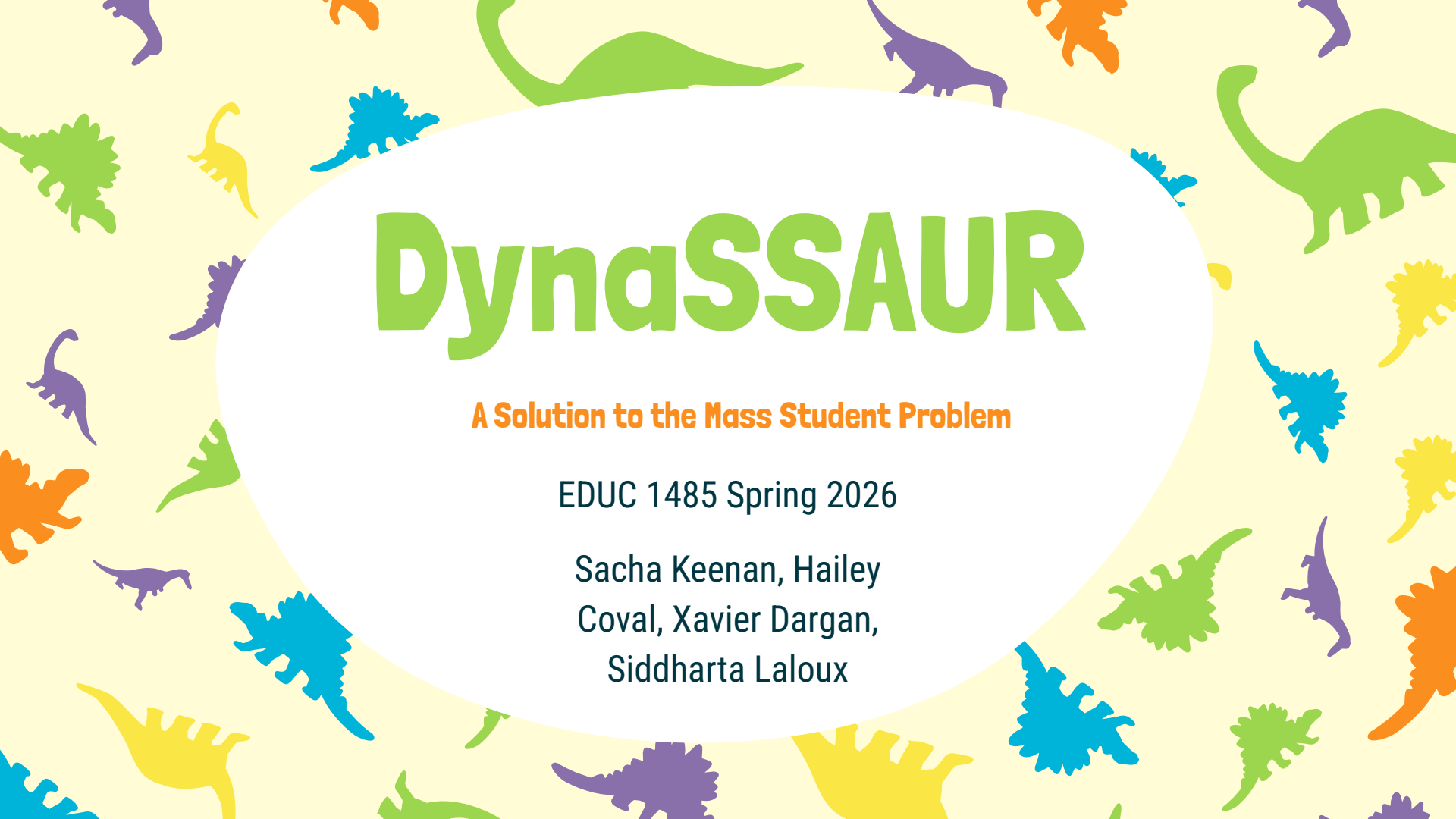




DynaSSAUR

A Solution to the Mass Student Problem

EDUC 1485 Spring 2026

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Meet the Team



Siddharthasaurus
Lalouxensis
The Paleontologist



Hailey Coval
The Poet



Siddharta Laloux
A Student



Santino Laroque
The Art Dealer



Sergio Valoux
The Ambassador



Simeon Laroche
The Photographer



Sacha Keenan
The Chemist



Sidario Leveau
The Finance



Sandro Delarue
The Archivist



Xavier Dargan
The Dinosaur Enthusiast



Salvador Laveau
The Diplomat

THE MASS STUDENT PROBLEM

Current pedagogical constraints prevent one-on-one attention, leading to large lecture styles that fail to foster discussion or hands-on learning.

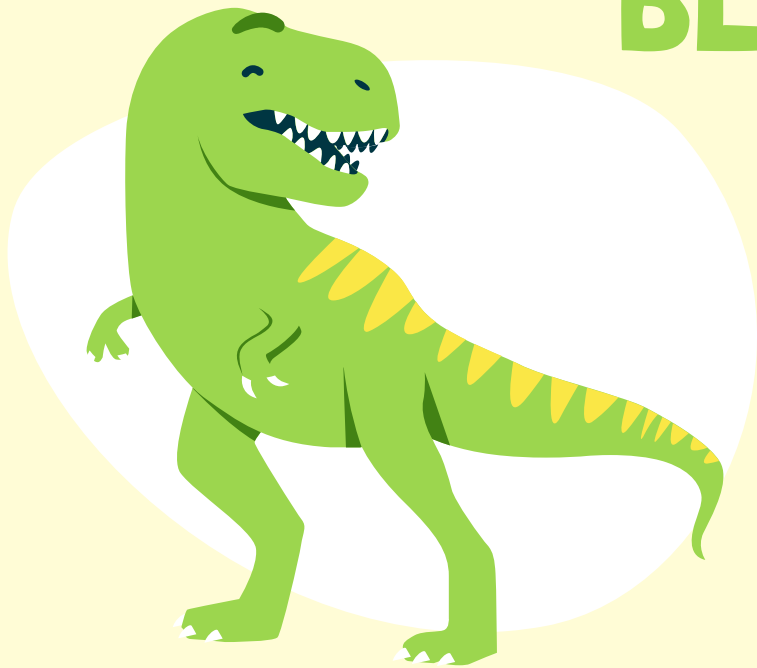
Demand vs. Capacity

Introductory CS courses are unable to keep up with the demand of large classes of students all asking different questions simultaneously.

The Scaling Issue

Scaling the necessary level of individualized tutoring to full university courses is logistically and financially impossible.

BLOOM'S 2 SIGMA PROBLEM



Bloom found that the average student receiving one-on-one tutoring performs two standard deviations above students in a conventional classroom—meaning the tutored student outperforms 98% of their peers.





Situated Cognition & Cognitive Apprenticeship

Knowledge cannot be separated from the activity and culture in which it is developed. A tutor does not just lecture; they model expert behavior and guide the novice through

(Brown, Collins, & Duguid, 1989)

The Failure

Conventional introductory CS classes fail to scale because they strip away the situated, guided practice of apprenticeship and replace it with mass-produced knowledge delivery which can also be applied to almost every other STEM field. Our web IDE tackles this scaling crisis directly.



DYNASSAUR's Principles of Operation



Coaching

Students receive hints and feedback when stuck



Scaffolding

The app creates a structured outline based on the assignment



Articulation

Students come up with their own plans and reasoning before implementation



Reflection

Comes naturally with the productive struggle



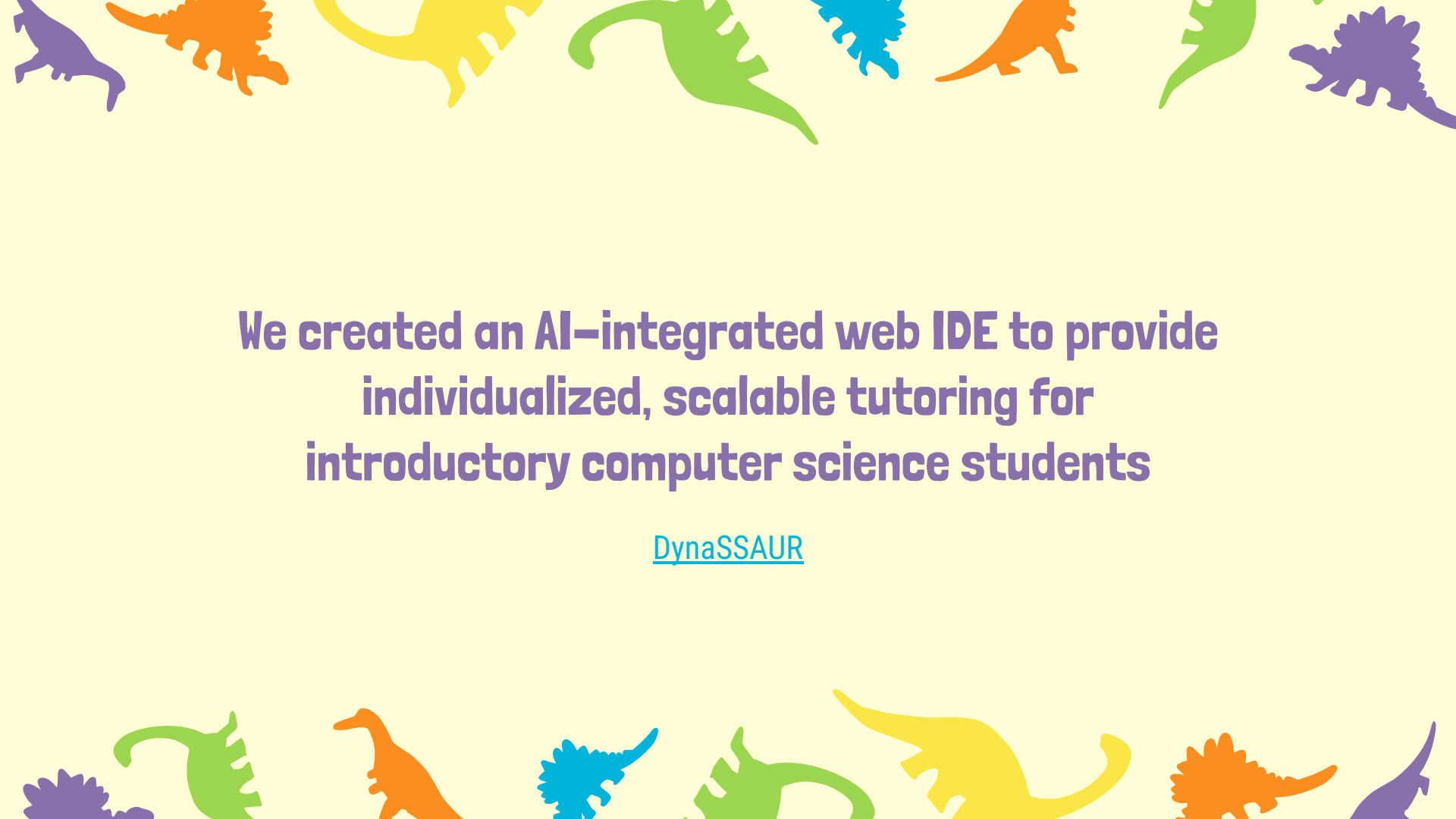
Modeling

Hints can often include example problems for the student to take after



Exploration

This model allows students to problem solve regardless of the auto grader



**We created an AI-integrated web IDE to provide
individualized, scalable tutoring for
introductory computer science students**

[DynaSSAUR](#)

Reconceptualizing Decomposition and Combating the Autograder

In CS, problem decomposition—the ability to break down complex tasks into well-defined components—is perceived as the most challenging programming skill for learners to master.

III-Structured Problem Solving (Jonassen, 1997)

CS education often tests students on "well-structured" problems (clear parameters, single correct answers), but real-world programming relies on solving "ill-structured" problems. Decomposition is the fundamental skill needed to navigate ambiguity and translate between the two



Creating Active Participators

Legitimate Peripheral Participation (Lave & Wenger, 1991)

Becoming a computer scientist requires enculturation into a "community of practice." TAs typically facilitate this by acting as cognitive masters to the student apprentices.

Sitting in a lecture hall does not demand attention from the students who fill it, and, with LLMs, nor do the assignments you give them, making practice illegitimate



WHAT MAKES US DIFFERENT

Unrestricted AI eliminates the friction of learning, bypassing the very mechanism required for neural pathways to develop.

We are restricted - we don't provide answers, we provide guidance

Current Use Causes and Issues

Loss of Productive Struggle

Task-Flation and Solutionism - (Morozov 2013)

Illusion of Understanding - (Bender et al. 2021)





Agency over Efficiency

POSSIBLE LIMITATIONS AND FURTHER THOUGHTS



HALLUCINATION

There's always a possibility the AI could get something wrong



REDUCING TA'S

That's not our goal nor what we think will occur



LLM PREVENTION

This app doesn't actually keep students from using an LLM



A decorative border at the top of the slide features a row of colorful dinosaur silhouettes in green, purple, orange, blue, yellow, and purple. A similar border is at the bottom with silhouettes in purple, orange, yellow, green, blue, orange, green, and purple.

THANK YOU FOR LISTENING!!!